

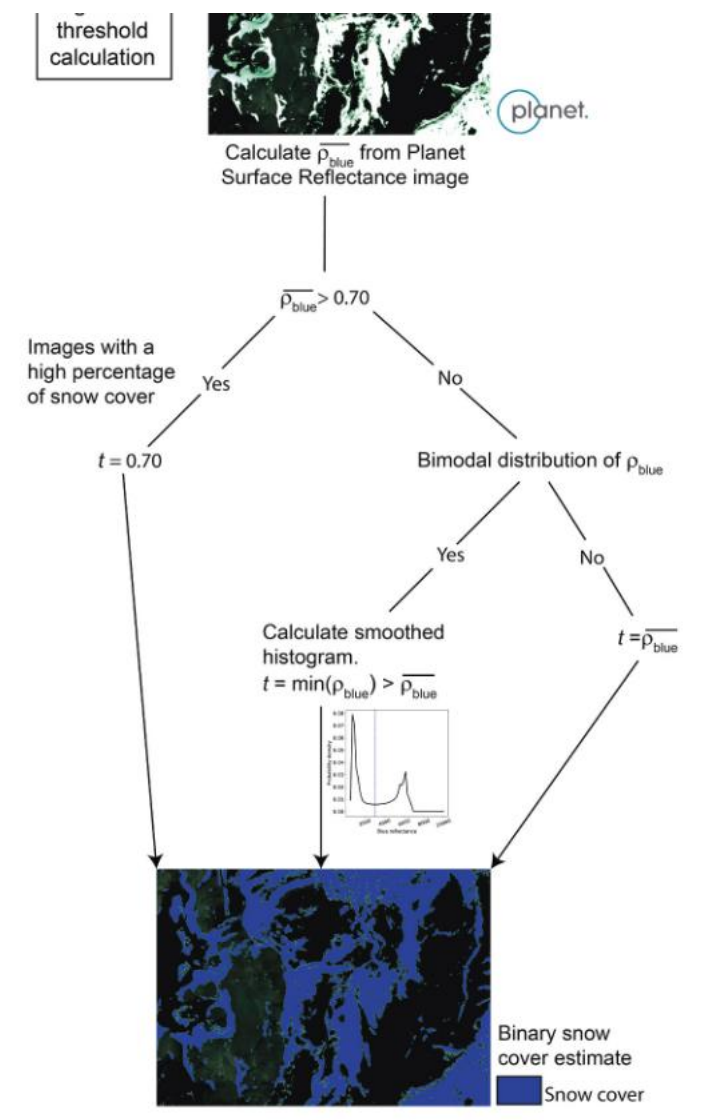
Blue Snow Threshold;

Problem: Planet Scope has no SWIR band, which is needed for the classic “NDSI: normalized difference snow index”.

Therefore, we used the approach of identifying snow patches as suggested in the following paper

Thaler, E. A., Crumley, R. L., and Bennett, K. E.: Estimating snow cover from high-resolution satellite imagery by thresholding blue wavelengths, Remote Sensing of Environment, 285, 113403, doi:10.1016/j.rse.2022.113403, 2023.

Blue Snow Threshold, based only on the reflection of the blue band and a scene individual threshold (see workflow from Thaler et. al. 2023 below)



As this is not directly available and they did it in python, I rebuild it equally in R.

Their results were validated like this:

- compared to airborne lidar surveys of SCA, the BST-derived SCA estimates

- F-scores between 0.81 and 0.94 in four study areas in California and Colorado.

Reference data like this was not available for Taylor valley

Therefore, and because visual comparison between using 1st min and mean_blue as threshold looked like the first could possibly be underestimating snow cover, for every PlanetScope scene a BST based prediction was calculated with both thresholds.

Afterwards a stratified random sample (10 points each for both snow, both not snow, and the differently assigned pixels) was gained for a buffer around each plot (p2, p4), where the buffer was built to not include glaciers or lakes. Stratified random sampling was chosen, because no snow covers way more area than snow in this area of Taylor Valley.

For these points, snow/ no snow was afterwards assigned manually in QGIS by visual interpretation of the corresponding True colour composite for each scene.

Please be aware that this method of GT sampling could lead to a bias because of wrong interpretations of the interpreter.

Observations while assigning:

- Differences in classification occur mostly on the edges of snow patches
- Small patches are overlooked by the original TH when more snow is present in general in the scene
- If there is less snow in the scene, the mean_blue TH leads to a high number of False positive snow
- For the time period is a melting of snow/ changing of patches present, based on the visual impression.

Afterwards the F-score was calculated for both situations with the F_meas function provided in the caret package.

- F_score original algorithm: 0.9 -> perfectly in line with findings from Thaler et.al 2022
- F_score_mean_blue: 0.71 (reflects impression during gaining GT)

Findings: BST can be used to identify snow patches in Taylor valley. The original approach will be used for further analysis/ predictor calculations.